Acceleration and Stability Regions of First-Order Methods on Strongly Convex Quadratics

Shuo Guo

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Abstract

Why accelerated methods are faster, and why they are often more fragile, is a basic conceptual questions in optimization. This paper studies that trade-off for gradient descent (GD), Heavy-Ball momentum (HB), and Nesterov acceleration gradient (NAG) on strongly convex quadratics. We show that acceleration comes from using more information: moving from one-state to two-state recursions and, in NAG, using a look-ahead gradient. On quadratics this yields faster contraction, but also pushes the dynamics closer to the stability boundary. We compare deterministic stability and noisy-gradient robustness, and distinguish boundedness, transient amplification, and steady-state robustness.

1 Introduction

Acceleration is a central idea in optimization, while Nesterov's acceleration is its most celebrated example. It achieves the optimal worst-case convergence rate while using only first-order information.[5]. Later developments connected accelerated methods to estimate sequences, Lyapunov functions, continuous-time dynamics, control-theoretic robustness, and noise-sensitive stochastic behavior [6, 10, 3, 1, 12]. Now we want to explain:

> Where does acceleration really come from, and is reduced stability an intrinsic price for obtaining it?

This paper addresses that question in a deliberately restricted setting: strongly convex quadratic objectives and three basic first-order methods, gradient descent (GD), Heavy-Ball momentum (HB), and Nesterov acceleration gradient (NAG). On quadratic objectives, the geometry is transparent; the Hessian eigen-directions decouple; the algorithms reduce to scalar linear recursions; and the relevant quantities — contraction, stability, oscillation, and noise amplification — can all be written explicitly.

The paper is motivated by three observations.

First, acceleration is not merely "adding a momentum term." On quadratics, it amounts to replacing a one-state linear recursion by a two-state recursion with more freedom to shape the decay of spectral modes. This is the mechanism behind the familiar improvement from the slowest-mode behavior of GD to the better worst-case behavior of accelerated schemes.

Second, that additional freedom is not free. Recent work on stability regions for gradient-based optimizers shows, in a deterministic quadratic setting, that certain Nesterov momentum admit a strictly smaller stability region than GD when stability is defined as boundedness of the recursion under a given learning-rate–curvature pair [2]. This suggests that acceleration may narrow the admissible parameter region.

Third, boundedness is only one notion of stability. Even when the recursion remains bounded, an accelerated two-state filter may display larger transient peaks, stronger oscillations, and worse noise amplification than GD. This is precisely the type of trade-off highlighted in the robust-control and inexact-oracle literatures [3, 1, 12].

The goal of this paper is to organize the discussion around one problem:

> On strongly convex quadratics, how does stronger information usage create acceleration, and why does the same mechanism often shrink the stability margin and worsen robustness to gradient uncertainty?

The rest of the paper is organized as follows. Section 2 defines the quadratic model and introduces three notions of stability. Section 3 explains where acceleration comes from. Section 4 derives deterministic stability regions for GD, HB, and NAG. Section 5 studies noisy gradients and transfer-function amplification. **??** discusses implications for machine learning. **??** briefly explains how the same questions extend to IQC analysis, estimate-sequence sensitivity, and accelerated second-order methods.

2 Problem setting and three notions of stability

2.1 Strongly convex quadratic model

We study minimization of

$$f(x) = \frac{1}{2} x^\top Q x, \tag{1}$$

where $Q \in \mathbb{R}^{d \times d}$ is symmetric positive definite. Let

$$0 < m = \lambda_{\min}(Q) \le \lambda_{\max}(Q) = L.$$

The unique minimizer is $x^\star = 0$.

Because Q is symmetric positive definite, it can be diagonalized with an orthonormal eigenbasis. In that basis, the dynamics decouple along the eigen-directions of Q, so that each mode evolves independently according to a scalar recursion with eigenvalue $\lambda_i \in [m, L]$. Hence the multivariate problem reduces exactly to the scalar quadratic model

$$\phi_i(z) = \frac{1}{2} \lambda_i z^2, \qquad \lambda_i \in [m, L]. \tag{2}$$

2.2 Algorithms under consideration

We compare three methods.

Gradient descent (GD). With stepsize $\eta > 0$,

$$x_{k+1} = x_k - \eta \nabla f(x_k). \tag{3}$$

Heavy-Ball momentum (HB). With stepsize $\eta > 0$ and momentum parameter $\beta \in [0, 1)$,

$$x_{k+1} = x_k - \eta \nabla f(x_k) + \beta (x_k - x_{k-1}). \tag{4}$$

Nesterov's accelerated gradient (NAG). We adopt the classical look-ahead form

$$y_k = x_k + \beta(x_k - x_{k-1}), \qquad x_{k+1} = y_k - \eta\nabla f(y_k). \tag{5}$$

2.3 Three notions of stability

We try to define *stability* with three notions.

Definition 1 (Deterministic boundedness stability)**.** *The sequence x_k is bounded and converges.*

Definition 2 (Transient sensitivity)**.** *While the sequence x_k converges, it will not have large overshoot or non-monotone peak.*

Definition 3 (Stochastic robustness)**.** *Under additive gradient noise, stochastic robustness refers to how strongly the recursion amplifies the noise in steady state.*

The first notion is the one underlying most closed-form stability-region results on deterministic quadratics [2]. The second and third notions are closer to what practitioners often mean when they say that a method is "fragile" or "hard to tune."

3 Where does acceleration come from?

3.1 From one-state to two-state dynamics

On the scalar quadratic (2), GD becomes

$$e_{k+1} = (1 - \eta\lambda)e_k, \tag{6}$$

where $e_k = x_k - x^\star = x_k$. This is a one-state linear system: its behavior on each mode is controlled by a single multiplier $1 - \eta\lambda$.

By contrast, HB becomes

$$e_{k+1} = (1 - \eta\lambda + \beta)e_k - \beta e_{k-1},^{1} \tag{7}$$

and NAG becomes

$$e_{k+1} = (1 - \eta\lambda)\big((1 + \beta)e_k - \beta e_{k-1}\big). \tag{8}$$

These are two-state recursions. Their state can be written as (e_k, e_{k-1}). The extra state stores information about past motion, and therefore gives the method more freedom than GD. Such extra freedom creates the possibility of substantially faster worst-case behavior than GD. NAG is the classical example.

3.2 Look-ahead gradients and predictive correction

NAG evaluates the gradient at the extrapolated point

$$y_k = x_k + \beta(x_k - x_{k-1}).$$

In this sense, NAG has a simple prediction–correction structure. On quadratic problems, this often gives a better trade-off between fast contraction and oscillatory behavior than the basic HB update.

The same point can be expressed in state-space language. GD is a one-state linear system, whereas HB and NAG are two-state systems. So acceleration is a change in the underlying dynamical structure of the iteration.

[1] Someone will ask why we simply use e_k and e_{k-1} to represent momentum, instead of using all of the history e. The answer is that e_k and e_{k-1} could contains all history information. Writing $v_k = x_k - x_{k-1}$, we have $v_{k+1} = \beta v_k - \eta\nabla f(x_k)$ and $x_{k+1} = x_k + v_{k+1}$. Thus the full past does not need to be stored explicitly.

3.3 Polynomial filtering on the spectrum

A second answer to the question "where does acceleration come from?" is spectral. On a quadratic,

$$e_k = p_k(Q)e_0,$$

for a matrix polynomial p_k. For GD, the polynomial has the simple form

$$p_k(t) = (1 - \eta t)^k.$$

For two-state methods, the polynomial is generated by a second-order recurrence and can approximate zero on the spectral interval $[m, L]$ more effectively in the worst direction. This is the polynomial-filter viewpoint behind Chebyshev-type acceleration and the standard condition-number improvements for first-order methods [6].

Thus acceleration is not mysterious: it is enhanced spectral shaping. The cost is that more aggressive spectral shaping typically creates a sharper trade-off with stability margin.

3.4 Continuous-time intuition

The continuous-time viewpoint of Su, Boyd, and Candès strengthens the same message. They showed that Nesterov acceleration is modeled by a second-order ODE with damping [10]. In that picture, GD corresponds to a first-order dissipative flow, while acceleration introduces inertia. Low damping can produce rapid progress, but it also produces oscillation and overshoot. Restart rules become natural because they re-inject damping when the inertial phase becomes too oscillatory.

The ODE viewpoint should not be mistaken for a replacement of the discrete analysis in this paper. The continuous-time model explains the *mechanism* of acceleration and oscillation. The discrete stability region, however, still depends on characteristic roots of the actual recursion. We now turn to that discrete analysis.

4 Deterministic stability regions on quadratics

In this section we compare GD, HB, and NAG under deterministic gradients. Since the quadratic model decouples along eigen-directions, it suffices to analyze the scalar mode $\lambda \in [m, L]$.

4.1 Gradient descent

The scalar recursion is (6). Its characteristic root is

$$r = 1 - \eta\lambda.$$

The standard unit-disk condition immediately yields the following.

Proposition 1 (GD stability)**.** *For the scalar mode $\lambda > 0$, the GD recursion is asymptotically stable if and only if*

$$0 < \eta\lambda < 2. \tag{9}$$

Equivalently, on the full quadratic objective with largest eigenvalue L, the method is stable whenever $0 < \eta L < 2$.

This is the familiar fact that GD is "one-parameter stable" in a very transparent way: there is only one root, and the stability region is a simple interval.

4.2 Heavy-Ball momentum

For HB, the scalar recursion (7) has characteristic polynomial

$$r^2 - (1 - \eta\lambda + \beta)r + \beta = 0. \tag{10}$$

We apply the Jury criterion for second-order polynomials.

Proposition 2 (HB stability)**.** *Assume $\beta \in [0, 1)$. Then the HB scalar recursion is asymptotically stable if and only if*

$$0 < \eta\lambda < 2(1 + \beta). \tag{11}$$

Hence on the full quadratic objective, a sufficient and necessary condition for stability in every mode is

$$0 < \eta L < 2(1 + \beta).$$

Proof. Write the polynomial in the form $r^2 + a_1 r + a_0$ with

$$a_1 = -(1 - \eta\lambda + \beta), \qquad a_0 = \beta.$$

The Jury conditions for all roots to lie in the open unit disk are

$$1 + a_1 + a_0 > 0, \qquad 1 - a_1 + a_0 > 0, \qquad 1 - a_0 > 0.$$

Substituting the coefficients gives

$$\eta\lambda > 0, \qquad 2 + 2\beta - \eta\lambda > 0, \qquad 1 - \beta > 0,$$

which are equivalent to (11) together with $\beta < 1$. □

Two comments are important. First, in the boundedness sense, HB admits a wider stepsize interval than GD for fixed $\beta > 0$. Second, this does *not* imply that HB is uniformly more robust. A larger boundedness region says nothing about transient peaks or sensitivity to noise.

4.3 Nesterov accelerated gradient

For NAG in the form (5), the scalar recursion is

$$e_{k+1} = (1 - \eta\lambda)((1 + \beta)e_k - \beta e_{k-1}),$$

with characteristic polynomial

$$r^2 - (1 - \eta\lambda)(1 + \beta)r + (1 - \eta\lambda)\beta = 0. \tag{12}$$

The Jury test again yields a closed-form stability interval.

Proposition 3 (NAG stability)**.** *Assume $\beta \in [0, 1)$. Then the NAG scalar recursion is asymptotically stable if and only if*

$$0 < \eta\lambda < \frac{2(1 + \beta)}{1 + 2\beta}. \tag{13}$$

Therefore, for every fixed $\beta > 0$, the admissible stepsize interval of this NAG realization is strictly smaller than that of GD, since

$$\frac{2(1 + \beta)}{1 + 2\beta} < 2.$$

Proof. Write (12) as $r^2 + a_1 r + a_0$ with

$$a_1 = -(1-\eta\lambda)(1+\beta), \qquad a_0 = (1-\eta\lambda)\beta.$$

The Jury conditions become

$$1 + a_1 + a_0 = \eta\lambda > 0,$$
$$1 - a_1 + a_0 = 1 + (1-\eta\lambda)(1+2\beta) > 0,$$
$$1 - a_0 = 1 - (1-\eta\lambda)\beta > 0.$$

The first inequality gives $\eta\lambda > 0$. The second gives

$$1 + (1-\eta\lambda)(1+2\beta) > 0 \quad \Longleftrightarrow \quad \eta\lambda < \frac{2(1+\beta)}{1+2\beta}.$$

The third is automatically satisfied for $\beta \in [0,1)$ and $\eta\lambda \geq 0$. Hence (13) is necessary and sufficient. □

4.4 Interpretation: why the stability margin narrows

The three propositions reveal the structural point of this paper. Going from GD to HB and NAG does not simply "make the method faster." It changes the order of the discrete system. In each eigen-direction, GD has one root, while HB and NAG have two roots. Those extra degrees of freedom make it possible to shape the contraction factor more aggressively, but they also make it easier for the roots to move close to or across the unit circle.

For the NAG realization above, the stepsize bound is more restrictive than for GD. This is fully consistent with recent boundedness-based stability-region calculations for Nesterov-type methods on quadratic objectives [2]. The point is not that all accelerated methods must always have a smaller deterministic stability region than GD in every parameterization. The point is that the structural route to acceleration — richer state, stronger spectral shaping, and lower effective damping — naturally creates narrower margins in many representative realizations.

5 Noise amplification and robustness

Deterministic boundedness is not the whole story. Practical optimization, especially in machine learning, rarely has exact gradients. Mini-batch stochasticity, numerical error, and model mismatch all inject uncertainty into the update. In this section we study how that uncertainty propagates.

5.1 Additive gradient-noise model

For the scalar mode, suppose the queried gradient is

$$g_k = \lambda e_k + w_k,$$

where (w_k) is a mean-zero white-noise sequence with variance σ^2. The noise enters the algorithms as follows.

GD:

$$e_{k+1} = (1-\eta\lambda)e_k - \eta w_k. \tag{14}$$

HB:

$$e_{k+1} = (1 - \eta\lambda + \beta)e_k - \beta e_{k-1} - \eta w_k. \tag{15}$$

NAG: For the look-ahead realization, with gradient sampled at y_k,

$$e_{k+1} = (1 - \eta\lambda)\big((1 + \beta)e_k - \beta e_{k-1}\big) - \eta w_k. \tag{16}$$

5.2 Transfer functions

Using the z-transform convention in which multiplication by z^{-1} corresponds to one-step delay, we obtain the noise-to-state transfer functions

$$H_{\mathrm{GD}}(z) = \frac{-\eta z^{-1}}{1 - (1 - \eta\lambda)z^{-1}}, \tag{17}$$

$$H_{\mathrm{HB}}(z) = \frac{-\eta z^{-1}}{1 - (1 - \eta\lambda + \beta)z^{-1} + \beta z^{-2}}, \tag{18}$$

$$H_{\mathrm{NAG}}(z) = \frac{-\eta z^{-1}}{1 - (1 - \eta\lambda)(1 + \beta)z^{-1} + (1 - \eta\lambda)\beta z^{-2}}. \tag{19}$$

These formulas show the essential distinction immediately. GD is a first-order IIR filter; HB and NAG are second-order IIR filters. Once the denominator is second-order, complex poles and resonance become possible. As the poles move closer to the unit circle, the filter becomes faster in a short transient sense but also more sensitive to specific frequencies of the noise.

5.3 Stationary variance and resonance

A convenient way to quantify stochastic robustness is via stationary variance. For a stable linear state-space model

$$s_{k+1} = As_k + Bw_k,$$

with w_k white and variance σ^2, the state covariance P solves the discrete Lyapunov equation

$$P = APA^\top + \sigma^2 BB^\top.$$

The steady-state mean-square error is a linear function of P. For GD, this can be computed explicitly from (14):

$$\mathrm{Var}(e_\infty) = \frac{\eta^2\sigma^2}{1 - (1 - \eta\lambda)^2} = \frac{\eta\sigma^2}{\lambda(2 - \eta\lambda)}. \tag{20}$$

The variance blows up as $\eta\lambda \uparrow 2$, which is exactly the deterministic stability boundary.

For HB and NAG, the analogous formulas are more cumbersome, but the transfer functions (18)–(19) already explain the phenomenon: second-order filters can develop a pronounced gain peak before the poles actually cross the unit circle. Thus two methods may both be deterministically stable while having very different stochastic robustness.

This observation aligns with the speed–robustness trade-off emphasized by Van Scoy and Lessard [12]. The sharper the transient contraction one attempts to obtain with a low-order two-state method, the larger the steady-state sensitivity one often has to accept under persistent noise.

5.4 Why boundedness is not enough

The previous section therefore needs to be read with care. It is meaningful to say that the HB boundedness interval is wider than the GD interval, and that the NAG interval in Proposition 3 is narrower than the GD interval. But these deterministic facts do not settle the practical notion of robustness.

The real lesson is that three questions must be separated:

(a) *Will the recursion stay bounded?*

(b) *How large can transient peaks become before convergence sets in?*

(c) *How much stationary error will persistent noise create?*

Accelerated methods may improve the first quantity in some parameterizations, worsen it in others, and still be clearly inferior in the third quantity. This is why speaking of "stability" without qualification is often misleading.

6 Conclusion

This paper studied acceleration and stability on the clean model of strongly convex quadratics. The main conclusions can be summarized as follows.

First, acceleration comes from stronger information usage. In first-order methods, this means moving from a one-state recursion to a two-state recursion, often together with look-ahead gradient evaluation. On quadratics this is equivalent to using a richer polynomial filter on the Hessian spectrum.

Second, the same mechanism that improves worst-mode contraction often narrows the stability margin. For the classical NAG realization analyzed here, the deterministic admissible stepsize interval is strictly smaller than that of GD. More generally, richer spectral shaping pushes the discrete dynamics closer to low-damping behavior.

Third, deterministic boundedness is only one notion of stability. In the presence of additive gradient noise, second-order filters exhibit resonance-like amplification that can increase steady-state error even when the recursion remains bounded.

The author is still curious about whether this analysis method could be applied to Adam, Nadam, and other optimizors [2]. Also, the author is thinking whether this article could deep our understanding to ML.s

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